

KONSTANTINA SOKRATOUS

Postdoctoral Fellow, University of Missouri

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Computational Cognitive Modeling | Simulation-Based Inference | Machine Psychometrics | Human and Artificial Intelligence

EDUCATION

Ph.D., Behavioral and Cognitive Neuroscience , University of Florida Dissertation: <i>Machine Learning Methods for Discovering Latent Structures: AI Tools for Cognitive Neuroscience</i> Advisor: Peter D. Kvam	2025
M.Sc., Behavioral and Cognitive Neuroscience , University of Florida Thesis: <i>SINs: Simulation Inversion Networks for Automated Model Fitting</i>	2022
M.Sc., Operations Management , Montpellier Business School, France Industry-funded master's program; rank: 2 of 16	2019
B.Sc. and M.Sc., Clinical Psychology , Université Paul Valéry Montpellier 3, France Licensed clinical psychologist in France	2018

ACADEMIC APPOINTMENTS

Postdoctoral Fellow, University of Missouri Department of Psychological Sciences; MizzouForward Artificial Intelligence Initiative; Columbia, Missouri	2025–Present
- Develop computational approaches for identifying the processes underlying decisions in humans and artificial agents, including simulation-based inference, generative latent-variable models, and machine psychometrics. - Serve as Co-PI or Co-I on four 2026 awards totaling \$136,000 for AI measurement, simulation-based inference, and human-work modeling. - Contributed amortized Bayesian inversion methods to an accepted IEEE DSAA paper on multi-contrast neutron reflectometry; now lead the project's second phase modeling expert decisions for autonomous experimentation. - Build multimodal computational frameworks connecting behavioral tasks, ecological momentary assessment, alcohol-administration data, neural measures, and self-report.	
Graduate Research Assistant and Doctoral Researcher, University of Florida Decision Dynamics and Modeling Laboratory; Advisor: Peter D. Kvam	2020–2025
- Joined as the laboratory's first doctoral student and helped establish its computational workflows, emerging research program, and practices for supporting subsequent student collaborators. - Developed simulation-inversion networks for amortized, likelihood-free estimation of computational cognitive models. - Built dynamic models of intertemporal choice, pricing, anticipatory timing, shifting alternatives, and resource allocation under uncertain deadlines. - Integrated response time and choice accuracy within cognitive process models, improving the reliability and validity of implicit-association measurement. - Developed variational-autoencoder approaches for discovering nonlinear latent behavioral structure and trajectories.	
Instructor of Record, University of Florida Department of Psychology; PSY 2012: General Psychology	2024
Designed and delivered the course independently; student evaluations exceeded department and college averages.	

PROFESSIONAL EXPERIENCE

Operations Analyst, Olea Medical / Canon Inc. La Ciotat, France	2018–2020
Applied statistical modeling, optimization, and data analysis to problems in automated medical diagnosis and human-algorithm decision-making.	
Clinical Psychology Fellow CMPEA Franco Basaglia and Institut de Cancérologie du Gard; Sète and Nîmes, France	2016–2018

PUBLICATIONS

Graduate and undergraduate student coauthors are marked with an asterisk.

Peer-Reviewed Journal Articles

1. Sokratous, K., & Kvam, P. D. (2026). Time flips and price dips: Modeling preference reversals between intertemporal choice and pricing.

- Hintze, A., **Sokratous, K.**, & Thomas, I. (2026). When language models date each other: Linguistic alignment, missing gender patterns, and RLHF preference collapse in AI-AI romantic interactions. *Computers in Human Behavior: Artificial Humans*.
- Davis-Stober, C. P., **Sokratous, K.**, & Vandekerckhove, J. (2026). Comment on Shiffrin et al. (2026): Illusions of replication, illusions of truth. *Computational Brain & Behavior*.
- Kvam, P. D., **Sokratous, K.**, Fitch, A. K.*, & Vassileva, J. (2025). Comparing likelihood-based and likelihood-free approaches to fitting and comparing models of intertemporal choice. *Behavior Research Methods*.
- Kvam, P. D., **Sokratous, K.**, & Fitch, A. K.* (2025). Decisions among shifting choice alternatives reveal option-general representations of evidence. *Psychological Review*.
- Kvam, P. D., **Sokratous, K.**, Fitch, A. K.*, & Hintze, A. (2024). Using artificial intelligence to fit, compare, evaluate, and discover computational models of decision behavior. *Decision*.
- Kvam, P. D., Irving, L. H.*, **Sokratous, K.**, & Smith, C. T. (2024). Improving the reliability and validity of the IAT with a dynamic model driven by similarity. *Behavior Research Methods*, 56.
- Sokratous, K.**, Fitch, A. K.*, & Kvam, P. D. (2023). How to ask twenty questions and win: Machine-learning tools for assessing preferences from small samples of willingness-to-pay prices. *Journal of Choice Modelling*, 48, 100418.
- Michel, A., [...], **Sokratous, K.**, & Blanc, N. (2019). Dessine-moi une réalité plus belle: La réalité virtuelle vue par les patientes atteintes d'un cancer du sein. *Psycho-Oncologie*, 13(1).

Peer-Reviewed Conference Proceedings

- Herrera, R. D., *Gollapelli, M.*, Roy, U.*, **Sokratous, K.**, Phan, M. D., Calyam, P., Davis-Stober, C. P., & Malik, T. (2026). Amortized Bayesian inversion of multi-contrast neutron reflectometry. *IEEE International Conference on Data Science and Advanced Analytics (DSAA 2026)*. **Accepted**.
- Sokratous, K.**, & Kvam, P. D. (2026). The role of temporal uncertainty in procrastination. *Proceedings of the Annual Conference of the Cognitive Science Society*. **Accepted**.
- Kvam, P. D., **Sokratous, K.**, [...], & Unruh, E.* (2021). Preference reversals between intertemporal choice and pricing. In T. Fitch, C. Lamm, H. Leder, & K. Teßmar-Raible (Eds.), *Proceedings of the 43rd Annual Conference of the Cognitive Science Society* (pp. 555–561).

Manuscripts Under Review and Preprints

- Sokratous, K.**, Fitch, A. K.*, Hintze, A., & Kvam, P. D. (2026). *The timing of suboptimality: How temporal uncertainty drives procrastination behavior*. Revision requested. **Preprint**
- Sokratous, K.**, Vandekerckhove, J., & Davis-Stober, C. P. (2026). *Machine psychometrics for artificial agents*. Under review. **Preprint**
- Ranjan, S., **Sokratous, K.**, & Makwana, M. (2026). *Psych Scanner: A framework for systematic cognitive evaluation of large language models*.
- Ranjan, S., Makwana, M., **Sokratous, K.**, & Odegaard, B. (2026). *metasignal: A Python package for comprehensive metacognitive analysis and decision-making*. Under review at the *Journal of Open Source Software*. **Preprint**
- Wang, Y., Fitch, A. K., **Sokratous, K.**, & Kvam, P. D. (2026). *Integrating gaze into models for multi-attribute pricing*. Under review. **Preprint**
- Ranjan, S., **Sokratous, K.**, & Odegaard, B. (2026). *Reality monitoring in large language models: Self-knowledge that transforms with conversation memory*. Under review. **Preprint**
- Hintze, A., **Sokratous, K.**, Fitch, A. K.*, & Kvam, P. D. (2026). *Large language models act like evolved agents in public-goods dilemmas*. Under review. **Preprint**
- Sokratous, K.**, Hawkins, G., & Kvam, P. D. (2026). *Why am I always late? Modeling the cognitive mechanisms underlying anticipatory timing under uncertainty*. **Preprint**
- Zhao, B., Radev, S. T., **Sokratous, K.**, & Kvam, P. D. (2026). *Moderational learning: A machine-learning framework for identifying individual differences in stimulus-response mappings*. Under review.
- Sokratous, K.**, Hintze, A., Fitch, A. K.*, & Kvam, P. D. (2026). *Procrastination in humans and machines*. **Preprint**

Invited Book Chapter

- Sokratous, K.**, & Kvam, P. D. (2026). Invited contribution to *The Cambridge Handbook of Judgment and Decision-Making*. In preparation.

Selected Work in Progress

- Sokratous, K.**, & Kvam, P. D. Nonlinear generative methods for the discovery of latent behavioral structure.
- Sokratous, K.**, Fitch, A. K., & Kvam, P. D. Variational autoencoders for the discovery of latent behavioral trajectories, with an application to the Berlin city data.
- Sokratous, K.**, Hintze, A., Fitch, A. K., & Kvam, P. D. Temporal uncertainty in optimal decision-making: A comparison between humans and machines.
- Sokratous, K.**, & Davis-Stober, C. P. Variational autoencoders for modeling upper-limb asymmetry in patients with neural atrophy.

RESEARCH FUNDING AND TECHNOLOGY TRANSFER

Skan.AI Private Company Gift

Human Work Automation and Context Extraction; Co-PI. Announcement

2026 | \$40,000

University of Missouri Big Idea Initiative

A New Measurement Science for the Evaluation of AI Systems; Co-PI

2026 | \$75,000

William K. Estes Fund <i>Unlocking New Horizons in Cognitive Modeling with Simulation-Based Inference; Co-I</i>	2026 \$20,000
MU Postdoctoral Association Research Grant <i>Toward a Theory of Machine Psychometrics</i>	2026 \$1,000
A. G. Leventis Foundation Educational Grant Three-year award; Sole PI	2022 \$36,000
University of Florida Informatics Institute AI Fellowship One of five fellowships awarded for the advancement of AI research	2021 \$45,000
Technology Licensing Agreement T18841, UF Innovate Automated cognitive-modeling system for assessing health-related behavior	2022

HONORS AND AWARDS

- **2026** — MUPA Peer Reviewed Travel Award, University of Missouri (\$500)
- **2024** — Gerber Award, University of Florida (\$1,000)
- **2024** — David Robert Ferguson Memorial Award, University of Florida (\$2,000)
- **2023** — Women of MathPsych Networking Award (\$550)
- **2023** — Grinter Fellowship for Extraordinary Recruits, University of Florida Graduate School (\$6,000)
- **2022** — Women Leads Fellowship, University of Florida
- **2022** — International Student Achievement Award, University of Florida International Center
- **2020–2025** — Graduate Fellowship, University of Florida (\$125,000)
- **2020–2025** — Travel Awards, University of Florida Department of Psychology, Graduate Student Council, and College of Liberal Arts and Sciences (\$7,000)
- **2016** — Postgraduate Scholarship, Republic of Cyprus (\$12,000)

INVITED TALKS

1. **Introduction to Generative AI for Model Discovery.** Summer School on Simulation-Based Inference, The Ohio State University, July 2026.
2. **Machine-Learning Methods for Model Fitting and Discovery.** Mellenbergh Lecture Series, University of Amsterdam, June 2025.
3. **Machine-Learning Methods for Model Estimation, Comparison, and Discovery.** Scripps Research Institute, San Diego, February 2025. Invited by Ann Kennedy.
4. **New Methods and Models in Decision-Making.** Decision Dynamics and Modeling Laboratory, Carnegie Mellon University, January 2025. Invited by Cleotilde Gonzalez.
5. **Machine-Learning Methods for Model Fitting and Discovery.** Shenhav Laboratory, University of California, Berkeley, December 2024.
6. **New Methods and Models in Decision-Making.** Laboratory for Neural Computation and Behavior, Brown University, December 2024. Invited by Michael Frank.
7. **AI as a Model of Behavior.** CoCoCo Laboratory and Enfascination, Cornell University / University of California, Davis, November 2024.
8. **Machine-Learning Methods for Model Fitting and Discovery.** Cognitive and Computational Neuroscience Laboratory, University of California, Berkeley, October 2024.
9. **Machine-Learning Methods for Advancing Scientific Inquiry.** Decision Psychology Colloquium, The Ohio State University, October 2024.
10. **Unveiling the Hidden: Machine-Learning Approaches for the Discovery of Latent Structures.** Deep Learning Symposium, MathPsych / International Conference on Cognitive Modeling, 2024.
11. **Machine-Learning Approaches to Estimating and Comparing Models of Intertemporal Choice.** Invited symposium in memory of A. J. J. Marley, MathPsych / International Conference on Cognitive Modeling, 2023.
12. **Dynamic Models of Decision-Making and Their Applications to Health-Relevant Behavior.** Center for Behavioral Economics in Health Research, University of Florida, 2022.
13. **Machine-Learning Approaches for Automated Model Fitting and Comparison.** University of Florida Informatics Institute, 2022.

CONFERENCE PRESENTATIONS

Talks

1. Sokratous, K. (2026). Machine learning for theory development in models of the mind. *Mind and Machine Alignment Summit*.
2. Sokratous, K. (2026). The role of temporal uncertainty in procrastination. *Annual Conference of the Cognitive Science Society*.
3. Kvam, P. D., Sokratous, K., Fitch, A. K.*, & Hintze, A. (2023). A computational theory of resource allocation and procrastination under uncertain time deadlines. *Psychonomic Society Annual Meeting*.
4. Sokratous, K., & Kvam, P. D. (2022). SInS: Simulation inversion networks for automated model fitting. *MathPsych / International Conference on Cognitive Modeling*.
5. Kvam, P. D., Sokratous, K., [...], & Unruh, E.* (2021). Preference reversals between intertemporal choice and pricing. *Annual Conference of the Cognitive Science Society*.

6. **Sokratous, K.**, Hawkins, G., & Kvam, P. D. (2021). A model of timing in simple anticipatory decisions. *MathPsych / International Conference on Cognitive Modeling*. Voted second most popular talk among more than 200 presentations.
7. Kvam, P. D., & **Sokratous, K.** (2021). Modeling preference reversals between intertemporal choice and pricing. *MathPsych / International Conference on Cognitive Modeling*. Voted sixth most popular talk.

Posters

1. **Sokratous, K.**, & Kvam, P. D. (2024). Procrastination: Humans versus machines. *Society for Judgment and Decision Making*.
2. **Sokratous, K.**, & Kvam, P. D. (2024). Machine-learning approaches for the discovery of latent structures. *Psychonomic Society Annual Meeting*.
3. **Sokratous, K.** (2021). Timing in simple anticipatory decisions. *Annual Conference of the Cognitive Science Society*.
4. Michel, A., [...], & **Sokratous, K.** (2018). La réalité virtuelle vue par les patientes atteintes d'un cancer du sein. *Association Française de Psychologie de la Santé*.
5. **Sokratous, K.** (2017). Quelle efficacité de la réalité virtuelle en cancérologie: Une revue de la littérature. *13e Journée GSO*.

TEACHING AND WORKSHOP LEADERSHIP

Instructor and Organizer

- **Unlocking New Horizons in Cognitive Modeling with Simulation-Based Inference**, Summer School Organizer and Instructor, The Ohio State University, 2026. [Course materials](#)
- **Mind and Machine Alignment Summit**, Workshop Organizer and Instructor: Machine-Learning Methods for Science, The Ohio State University, 2026.
- **PSY 1012: General Psychology**, Instructor of Record, University of Florida, 2024. Student evaluations exceeded department and college averages.
- **Clinical Psychology Laboratory**, Université Paul Valéry Montpellier 3, 2017.

Teaching Assistant

- **EXP 3604: Cognitive Psychology**, University of Florida, Fall 2023. Online synchronous course; 432 students.
- **PSY 3213L: Research Methods in Psychology**, University of Florida, Fall 2022. Online synchronous course; 362 students.

TECHNICAL EXPERTISE

Methods: Computational cognitive modeling; evidence-accumulation and diffusion models; simulation-based inference; amortized Bayesian inference; variational autoencoders and deep generative models; joint modeling of behavioral, neural, and intensive longitudinal data; psychometrics; machine learning

Programming: Python, R, MATLAB, LaTeX; JavaScript and HTML

Languages: Greek, English, French, Spanish

SERVICE AND LEADERSHIP

- **2026–Present** — Vice President, Missouri Postdoctoral Association, University of Missouri
- **2020–2025** — Bargaining Committee Member and Organizer, Graduate Assistants United, University of Florida
- **2024** — Poster Judge, Psychology Graduate Student Organization Undergraduate Poster Forum, University of Florida
- **2024** — Poster Judge, Junior Science and Humanities Symposium, University of Florida
- **2022–2024** — Elected Senator, United Faculty of Florida
- **2022–2024** — Elected Delegate, Florida Education Association

Reviewing

Psychological Review; Computational Brain & Behavior; Nature Computational Science; Proceedings of the Annual Conference of the Cognitive Science Society; Journal of Mathematical Psychology; Behavior Research Methods